

Cape Cod Method (Stanard-Bühlmann)

An actuary at the company has the following experience for *cumulative paid claims* and premiums for the comprehensive motor insurance line. The CDFs are cumulative age-to-ultimate factors derived from the development triangle.

Accident year	Earned premium (M COP)	Reported claims (M COP)	CDF (age-to-ult.)
2020	1,000	820	1.04
2021	1,100	790	1.12
2022	1,250	810	1.30
2023	1,400	680	1.72
2024	1,550	490	2.80

Key Formula — Cape Cod ELR

$$\text{ELR}_{\text{CC}} = \frac{\sum_j C_{j,\text{dev}}}{\sum_i (P_i \rho_i)}, \quad \text{where } \rho_i = \frac{1}{\text{CDF}_i}.$$

$$\text{IBNR}_{\text{CC},i} = q_i \text{ELR}_{\text{CC}} P_i, \quad \text{where } q_i = 1 - \rho_i.$$

- Calculate the **percentage developed** $\rho_i = 1/\text{CDF}_i$ for each accident year.
- Calculate the portfolio's **Cape Cod ELR**. Show the numerator and denominator explicitly.
- Calculate the **Cape Cod IBNR** for each year and the portfolio total. Present the results in a table.
- For 2024, compare the Cape Cod IBNR with the **Chain Ladder** IBNR. Determine the implied ELR that CL projects for that year and **interpret the difference** from an actuarial perspective.
- The approved **pricing** ELR was 76%. What three actuarial signals would the calculated Cape Cod ELR provide relative to this value? (brief answer, 3 points).

Answers

a) Percentage Developed

Year	CDF	$\rho_i = 1/\text{CDF}_i$	Percentage	$q_i = 1 - \rho_i$
2020	1.04	0.9615	96.15%	0.0385
2021	1.12	0.8929	89.29%	0.1071
2022	1.30	0.7692	76.92%	0.2308
2023	1.72	0.5814	58.14%	0.4186
2024	2.80	0.3571	35.71%	0.6429

Only 35.7% of the losses for 2024 have developed — the portfolio's largest source of uncertainty.

b) Portfolio Cape Cod ELR

$$\begin{aligned}
 \text{Numerator} &= \sum_i C_{i,\text{dev}} \\
 &= 820 + 790 + 810 + 680 + 490 \\
 &= 3,590 \text{ M COP.}
 \end{aligned}$$

The denominator is $\sum_i (P_i \rho_i)$:

Year	$P_i \rho_i$	Result (M COP)
2020	$1,000 \times 0.9615$	961.5
2021	$1,100 \times 0.8929$	982.2
2022	$1,250 \times 0.7692$	961.5
2023	$1,400 \times 0.5814$	814.0
2024	$1,550 \times 0.3571$	553.5
Total		4,272.7

$$\text{ELR}_{\text{CC}} = \frac{3,590}{4,272.7} \approx 84.04\%$$

The CC ELR of 84.04% is the portfolio's *self-consistent* expected loss ratio, weighted by the volume effectively observed at each stage of development.

c) Cape Cod IBNR by Year and Portfolio Total

$$\text{IBNR}_{\text{CC},i} = q_i \times 84.04\% \times P_i.$$

Year	q_i	ELR $\times P_i$ (M COP)	IBNR _{CC} (M COP)
2020	0.0385	840.4	32.4
2021	0.1071	924.4	99.0
2022	0.2308	1,050.5	242.5
2023	0.4186	1,176.6	492.5
2024	0.6429	1,302.6	837.5
Total	—	—	1,703.9

d) Comparison with Chain Ladder for 2024

Chain Ladder — Year 2024:

$$\begin{aligned}\text{Ultimate}_{\text{CL}} &= 490 \times 2.80 = \mathbf{1,372} \text{ M COP}, \\ \text{IBNR}_{\text{CL}} &= 1,372 - 490 = \mathbf{882} \text{ M COP}.\end{aligned}$$

Cape Cod — Year 2024:

$$\text{IBNR}_{\text{CC}} = 837.5 \text{ M COP}.$$

$$\begin{aligned}\text{Difference} &= \text{IBNR}_{\text{CL}} - \text{IBNR}_{\text{CC}} \\ &= 882 - 837.5 = +\mathbf{44.5} \text{ M COP} \quad (\text{CL} > \text{CC}).\end{aligned}$$

Implied CL ELR for 2024:

$$\frac{1,372}{1,550} \approx \mathbf{88.5\%} \quad \text{vs.} \quad \text{ELR}_{\text{CC}} = 84.04\%.$$

Interpretation: CL scales up the 490 M observed in 2024 by the CDF of 2.80, projecting an implied loss ratio of 88.5%. CC, in contrast, anchors the projection to the portfolio ELR (84.04%), resulting in a reserve that is 44.5 M lower. If the low loss ratio for 2024 reflects *normal administrative delays* (rather than a genuinely favorable year), CC is more stable. If there is a real emerging deterioration in 2024, CL will capture it sooner as the data mature.

e) Actuarial Signals Relative to the Pricing ELR

With $\text{ELR}_{\text{CC}} = 84.04\%$ versus a pricing ELR of 76%, there are three actuarial signals:

- **The emerging loss ratio exceeds the pricing assumption:** the portfolio's loss ratio is 8 percentage points higher than the pricing actuary anticipated. If this is a sustained trend, rates need to increase or underwriting needs to be strengthened.
- **Possible deterioration in the line of business:** this may reflect claims inflation (medical CPI, motor vehicle spare parts), a shift in the portfolio mix toward riskier segments, or a soft market cycle in which risks were accepted below technical rates.
- **The prior assumption in the classical BF method is outdated:** if 76% continued to be used as the ELR in Bornhuetter-Ferguson, reserves for recent accident years would be underestimated. CC provides an objective warning signal to update the prior assumption.

Rate on Line (ROL) and XL Burning Cost

The insurer structures an excess-of-loss (XL) reinsurance program with two layers. The **historical experience for Layer 1** (2,000M xs 1,000M) is as follows:

XL Program Structure

Range (M COP)	Level	Coverage	ROL
3,000–6,000	Layer 2 senior	3,000M xs 3,000M	5.0%
1,000–3,000	Layer 1 working	2,000M xs 1,000M	?
0–1,000	Retention	Cedant's retention	—

Year	Losses in layer (M COP)	Total direct premium (M COP)
2019	0	13,500
2020	1,200	14,200
2021	0	14,800
2022	0	15,500
2023	600	16,200
2024	0	17,000

Key Formulas — ROL and Burning Cost

$$\text{Burning Cost \%} = \frac{\text{Average annual losses in layer}}{\text{Average annual direct premium}},$$

$$\text{XL premium} = \text{Pure Burning Cost} \times (1 + \text{loading}),$$

$$\text{ROL} = \frac{\text{XL premium}}{\text{Layer limit}},$$

$$\text{Payback} = \frac{1}{\text{ROL}}.$$

The payback represents the implied recurrence period, in years, for a total loss.

- Calculate the **pure Burning Cost** for Layer 1 as a percentage of direct premium and as an absolute amount (M COP/year).
- Applying a **40% loading** to the pure Burning Cost, calculate the annual reinsurance premium and the **ROL** for Layer 1.
- Interpret the calculated ROL in terms of the **payback period**: how many years would the reinsurer expect to wait between losses exhausting the full limit of Layer 1?
- Layer 2 (3,000M xs 3,000M) has a ROL of 5.0%. Calculate its **reinsurance premium** and **payback period**. Compare it with Layer 1 and explain why the ROL is systematically lower for higher layers.
- If the reinsurer requires a **minimum ROL of 15%** for any working layer, what adjustment does this imply for Layer 1? How would this affect the program designer?

Answers

a) Pure Burning Cost for Layer 1

Total losses in layer (6 years):

$$0 + 1,200 + 0 + 0 + 600 + 0 = 1,800 \text{ M COP},$$

$$\text{Average annual losses} = \frac{1,800}{6} = \mathbf{300} \text{ M COP/year}.$$

Average annual direct premium:

$$\begin{aligned} \bar{P} &= \frac{13,500 + 14,200 + 14,800 + 15,500 + 16,200 + 17,000}{6} \\ &= \frac{91,200}{6} = 15,200 \text{ M COP/year}. \end{aligned}$$

$\text{Burning Cost \%} = \frac{300}{15,200} \approx 1.97\% \text{ of direct premium}$
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Only 2 of the 6 years had losses in the layer — low frequency but significant severities (1,200M and 600M). Historical Burning Cost is no guarantee of future experience: a single catastrophe could triple it.

b) XL Premium and ROL for Layer 1

XL premium (pure Burning Cost $\times$ 1.40):

$$300 \text{ M COP/year} \times 1.40 = \mathbf{420} \text{ M COP/year},$$

$$\text{Layer 1 limit} = 2,000 \text{ M COP},$$

$\text{Layer 1 ROL} = \frac{420}{2,000} = 21.0\%.$
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The 40% loading covers the reinsurer's expenses (commissions, administration) and profit margin. In practice, the loading ranges from 25% to 60% depending on the layer's volatility and the reinsurance market cycle.

c) Payback Period

$\text{Payback period} = \frac{1}{\text{ROL}} = \frac{1}{0.210} \approx 4.76 \text{ years}$

Interpretation: the reinsurer expects to cover the full limit (2,000 M COP) approximately once every 5 years.

In other words, the implied annual frequency of a total loss to the layer is 21%, equivalent to a “full limit” event every 5 years.

The payback period is a key metric for the program designer to assess whether the price is reasonable. If the technical actuary estimates that the layer will be exhausted every 3 years, a ROL of 21% (payback of 5 years) would *underprice* the risk.

d) Premium, ROL, and Payback for Layer 2

Layer 2 (3,000M xs 3,000M):

$$\begin{aligned}\text{Premium} &= 150 \text{ M COP}, \\ \text{Limit} &= 3,000 \text{ M COP}, \\ \text{Layer 2 ROL} &= \frac{150}{3,000} = \mathbf{5.0\%}, \\ \text{Layer 2 payback} &= \frac{1}{0.050} = \mathbf{20 \text{ years}}.\end{aligned}$$

Comparison:

Layer	ROL	Payback
Layer 1 (working)	21.0%	4.8 years
Layer 2 (senior)	5.0%	20 years

Why does the ROL decrease for higher layers? Layer 2 is triggered only when the total loss from a claim (or the aggregate loss over the period) exceeds 3,000 M COP, an attachment point much higher than that of Layer 1 (1,000 M). The probability of this happening is significantly lower — the expected loss to Layer 2 is small relative to its 3,000 M limit, so the reinsurer charges less in terms of ROL. In a well-structured XL program, the ROL decreases monotonically as the attachment point increases.

e) Minimum Required ROL and Implications

$$\begin{aligned}\text{Minimum required ROL} &= 15\%, \\ \text{Calculated ROL} &= 21.0\%.\end{aligned}$$

Layer 1 **already exceeds the minimum** ($21\% > 15\%$), so **no adjustment is required** in this case.

However, if the reinsurer required $\geq 21\%$ (an exact match), the program designer would have no room for negotiation.

If the BC had produced a ROL of 12% (alternative example):

$$\begin{aligned}\text{Minimum premium} &= 15\% \times 2,000 = 300 \text{ M COP/year}, \\ \text{Technical premium} &= 12\% \times 2,000 = 240 \text{ M COP}, \\ \text{Shortfall} &= 60 \text{ M}.\end{aligned}$$

Implications for the program designer when the ROL is insufficient:

1. **Raise the attachment point** of the layer (reduce the reinsurer's exposure).
2. **Reduce the layer limit** so that the premium represents a higher percentage of the limit.
3. **Renegotiate the loading** or expand co-reinsurance.
4. Supplement with market benchmark data if the company's own historical experience is limited (only 2 events in 6 years — low statistical credibility).